

Geometry
Unit 8
Lesson 6 Practice

Name **Answer Key** _____
Date _____
Period _____

Two congruent angles of an isosceles trapezoid have measures $(8x + 14)^\circ$ and $(12x - 8)^\circ$.

1. What is the value of x ?

$$\begin{aligned}8x + 14 &= 12x - 8 \\4x &= 22 \\x &= \frac{11}{2}\end{aligned}$$

2. What are the measures of each angle in the trapezoid?

$$\begin{aligned}8\left(\frac{11}{2}\right) + 14 &= 58^\circ \\180 - 58 &= 122^\circ\end{aligned}$$

For each quadrilateral description, select all shapes that match the description.

3. Quadrilateral $GRMW$ has two pairs of congruent sides and consecutive angles are supplementary.

- ☒ Parallelogram
- ☒ Rectangle
- ☒ Square
- ☒ Rhombus
- ☐ Kite

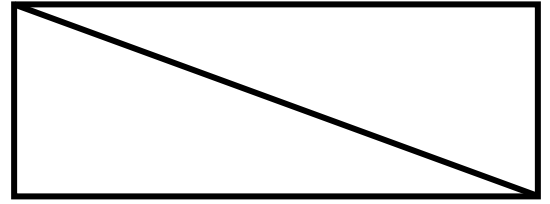
4. Quadrilateral $BNTV$ has four congruent sides and diagonals are perpendicular to each other.

- ☐ Parallelogram
- ☐ Rectangle
- ☒ Square
- ☒ Rhombus
- ☐ Kite

5. Quadrilateral $KPSX$ has two pairs of congruent sides and one pair of opposite congruent angles.

- ☐ Parallelogram
- ☐ Rectangle
- ☐ Square
- ☐ Rhombus
- ☒ Kite

Barrett is looking at a building on the school grounds which is shaped like a rectangle as shown. He knows the building has a height of $32x$ feet, width of $46x$ feet and the diagonal of the building is 200 feet.



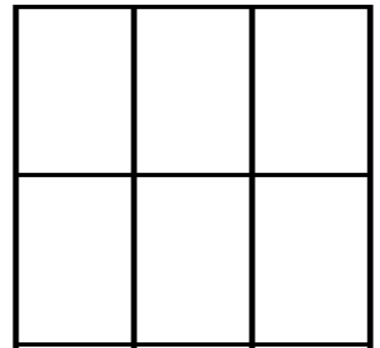
6. What is the value of x ? Round answers to the nearest hundredth.

$$\begin{aligned}(32x)^2 + (46x)^2 &= 200^2 \\ 3140x^2 &= 40000 \\ x^2 &= 12.739 \\ x &\approx 3.5692\end{aligned}$$

7. What is the height of the building? Round answers to the nearest hundredth.
height $\approx 32(3.5692) \approx 114.21$ ft

8. What is the width of the building? Round answers to the nearest hundredth.
width $\approx 46(3.5692) \approx 164.18$ ft

A wall paneling with a total area of 18 square feet is made up of 6 rectangles, as shown. There are two congruent rectangles along the height and three across the width. Each rectangle in the paneling is $(3x + 1)$ feet in height and $(2x - 1)$ feet in width.



9. What is the value of x ? Round your answer to the nearest hundredth.

$$\begin{aligned}18 &= 6[(2x - 1)(3x + 1)] \\ x &\approx 0.904\end{aligned}$$

10. What are the dimensions of the portion of the wall covered by the paneling?

$$\begin{aligned}\text{height of 1 panel} &= 3(0.904) + 1 \\ \text{height of 2 panels} &= 2[3(0.904) + 1] \\ \text{height} &\approx 7.424 \text{ ft}\end{aligned}$$

$$\begin{aligned}\text{width of 1 panel} &= 2(0.904) - 1 \\ \text{width of 3 panels} &= 3[2(0.904) - 1] \\ \text{width} &\approx 2.424 \text{ ft}\end{aligned}$$